

What drives urban carbon emission efficiency? – Spatial analysis based on nighttime light data

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HIGHLIGHTS

- The factors that affecting urban carbon emissions efficiency (CEE) are explored.
- The carbon emissions of 282 cities are estimated with the aid of nighttime light data.
- Urban CEE has significant and positive spatial spillover effects.
- Industrial agglomeration and collaboration are key to green production efficiency.

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ABSTRACT

This paper explores the factors that improve the efficiency of urban carbon emissions. Based on nighttime light data, the carbon emissions of 282 cities in China from 2004 to 2018 are estimated, and the carbon emission efficiency (CEE) is measured. The spatial autocorrelation results show that CEE in Chinese cities has significant and positive spatial spillover effects. The LISA low-low type cluster area of CEE presents an evolutionary trend of westward migration, and eventually formed two low-efficiency centers in the Central Plains city group and the Sichuan-Chongqing city group. The empirical results of the spatial Durbin error model show that urban population expansion, economic growth and R&D investment have effectively improved CEE, but industrial scale and agglomeration adversely affect CEE. In addition, human capital investment and foreign investment have not exerted their due green growth effects. Manufacturing agglomeration had a negative impact on local CEE, but had a warning effect on neighboring regions. Different from the manufacturing industry, the agglomeration of producer services industry can bring significant improvement to CEE, accompanied by a demonstration effect which can promote the CEE in adjacent regions. Further, this paper finds that developing the synergy between manufacturing and producer services industries can significantly restrain the negative effects of industry on CEE. This paper enriches the performance evaluation framework of low-carbon city development in theory, and varies the situations that previous researches only concerned about carbon emissions while ignoring emissions efficiency. This paper is a beneficial policy practice of the coordinated development of economic growth and ecological environmental protection in the regional coordinated development mechanism, which indicates that strengthening industrial collaboration can effectively improve the efficiency of green production in cities.

1. Introduction

Urban economic development and the control of greenhouse effect are two major problems confronted by developing countries [1]. Cities are the basic units of environmental policy implementation and the key areas for emission reduction. The realization of low-carbon emission

reduction targets cannot rely solely on end-of-pipe governance measures in the industrial sector. In-depth adjustments to urban layout, energy structure, and industrial structure must be carried out. This requires active cooperation among various sectors [2]. In the future, the emission reduction potential of the industrial sector will gradually decrease, while the energy demand of urban residents and the transportation

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sector will continue to rise [3]. Studies have shown that China's carbon emissions rebounded in 2017–2019, ending the steady decline in 2013–2016 [4]. China's total energy demand will continue to rise. The energy transition and realization of the carbon neutral goal will face greater pressure [5]. Under the constraints of low-carbon development, preventing the rebound risk of carbon emissions, realizing total energy demand control and low-carbon transformation of various sectors are urgent issues that need to be resolved in the healthy development of cities.

The performance measurement of carbon emissions is an important work in the field of environmental economics. Existing studies can be roughly divided into two categories: single-factor and total-factor carbon emission performance. The single-factor method mainly uses indicators such as carbon emissions per unit of GDP (i.e. carbon intensity) and energy intensity to measure carbon emission performance [6]. However, such simple indicators are difficult to measure the efficiency of the entire complicated urban system. Therefore, measuring carbon emission efficiency (CEE) from the perspective of total factor productivity has become the mainstream practice in academia [2]. The total-factor CEE not only reflects the cost of carbon emission, but also reflects the coupling relationship between environment and social development. The production frontier is constructed through the input–output relationship, and CEE is defined as the deviation of actual emissions from the theoretical optimal value on the frontier [3,7]. The increase in CEE reflects the optimization of productivity and the positive externalities of environmental protection, which means a better coordination between economic growth and carbon emission reduction [8]. This caters to the long-term needs of many countries, especially developing countries, to cope with climate change and transform their economic growth patterns [9]. It is believed that a region's carbon emissions are determined by geographic location, factor endowments, and historical factors. While the efficiency of carbon emissions is a comprehensive manifestation of urban economic decision-making and environmental behavior. Therefore, compared with total carbon emissions, this paper focuses more on the urban CEE.

Previous literature has basically confirmed the relationship between carbon dioxide emissions and urbanization, but few research has comprehensively studied the impact of various subsystems in the city on CEE. IPAT model shows that population, affluence and technology are the main factors affecting environmental quality [10]. Due to the concentration of a large number of population and production activities, the increase in housing, transportation and electricity demand has led to a large amount of urban carbon emissions. Cities are considered to be an important area for accomplishing energy conservation and emission reduction tasks [11]. Many scholars have discovered the EKC relationship between urbanization and carbon emissions, and this non-linear relationship is reflected through a variety of mechanisms, including population, economy, technology, etc [12]. Though the concentration of urban population enables the intensive use of resources, the resident consumption effect brought about by the expansion of the population makes cities the main source of carbon emissions, even exceeding the industrial sector [13]. Technological progress is an eternal topic in development economics, but the relationship between innovation and environmental quality is also unclear. The rebound effect of carbon emissions is considered to be a major obstacle to achieving the goal of carbon neutrality, which means that the expected energy savings are mostly offset by the additional energy consumption brought about by technological progress. Shao et al. [14] found that technological progress in resource-dependent cities will cause an energy rebound effect, leading to a rebound in carbon emissions.

In addition, regional heterogeneity and spatial correlation are also very significant features that affect regional environmental quality. Taking into account the spillover effects of environmental products [15], spatial econometric models have been widely used to explore the driving force behind carbon dioxide emissions [16]. Regional collaboration is a new direction for tackling climate change issues [8]. Many studies have

adopted spatial dynamic model analysis and found that technological progress in neighboring areas played an important role in reducing carbon emissions [2,17]. Due to the spatial interaction, urban eco-efficiency will inevitably have a diffusion or polarization effect on the surrounding areas [18]. CEE is also expected to cause such specific effects. China is the largest carbon emitter and has the largest population in the world. Research on China's urban carbon emission efficiency will provide a great reference for global climate governance.

The contribution of this paper is mainly reflected in three aspects. In terms of data sources, this paper is the first to study urban carbon emission efficiency (CEE) based on nighttime light data. Due to the lack of official carbon emission data sources, existing studies use alternative indicators such as electricity consumption and energy intensity to estimate carbon emissions [7,19]. Different types of energy and emission coefficients has led to large differences in the measurement of urban carbon emissions, which make the research results lacking comparability. Satellite remote sensing data has been used more and more widely in the fields of environmental economics [16,20]. Compared with the ground point-based monitoring data, satellite remote sensing data enhances the data consistency and comparability between regions over the long-term series [21]. This paper uses global NPP-VIIRS-like nighttime light data [22] from a cross-sensor calibration to estimate urban carbon emissions, and calculates urban CEE.

Secondly, in terms of model selection, this paper uses the spatial panel model to investigate the efficiency of urban carbon emissions. The first law of geography points out that there are spatial interactions and spatial spillover effects of geographic factors in economic and social activities, that is, spatial dependence. Unlike previous studies at the provincial or national level, such spatial dependence will be more significant at the city level. Ignoring the interaction factors between regions may lead to missed variable biases, thereby causing the endogenous problem of the results [2,23]. In the past ten years, China has built and improved its transportation infrastructure on a large scale. Neighboring regions are more likely to establish close cooperative relations and imitate each other's development mode [17]. Therefore, this paper adopts the method of exploratory spatial data analysis (ESDA) to test the spatial dependence of CEE on the global and local scale, and analyzes its changing trends in the spatial and temporal dimensions.

Existing researches have verified the non-linear relationship between carbon emissions and urbanization from multiple perspectives, such as energy system, technological progress, transportation demand, etc [24,25]. Some other scholars have studied the carbon emission performance of individual typical cities or regions [11,14]. However, the systematic analysis on the carbon emission efficiency of the entire city system, though complex and uncertain, is worthy of further study. Hence, the third contribution of this paper is to decompose and expand the influencing factors of the STIRPAT model, reflecting the nonlinear relationship between different factors and carbon emissions in the existing literature. For example, the population factor is decomposed into population size, population density and human capital level, and technical variables are decomposed into input variables and performance variables. In addition, in order to alleviate the negative effects of population and industry agglomeration, this paper also studies the mechanism of improving CEE from the perspective of regional openness and industrial collaboration.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on the influencing factors of carbon emission efficiency. Section 3 introduces the research methods of this paper, including the estimation of urban carbon emissions, the measurement of CEE and the setting of spatial panel models. Section 4 reports the empirical results and discusses the factors affecting CEE. The conclusions are discussed in the Section 5.

2. Literature review

Literature about the relationship between urban population

agglomeration and carbon emissions can be divided into three categories. One view is that the consumption effect of urban residents increases energy demand, which leads to an increase in carbon emissions. Cities encompass more than half of the world's population, occupy 2% of the earth's area, but generate more than 75% of the world's carbon dioxide emissions [12]. The consumption effect can be decomposed into an increase in population or in per capita income. Wang et al. [26] confirmed that the size of the population is the main reason for the increase of carbon emissions. Rafiq et al. [27] further found that population density and affluence can also increase carbon dioxide emissions and energy intensity. In addition to the carbon emissions caused by the residential sector, the expansion of cities will inevitably lead to an increase in other sectors. For example, the increase in transportation activities increases energy consumption and reduces energy efficiency [28,29]. Another strand of literature holds that the increase in population density can cause intensive effects, such as saving space to facilitate central heating and power supply, increasing the utilization rate of public transportation, sharing urban infrastructure, etc., reducing residential energy consumption, thereby reducing carbon emissions [30]. Other studies have combined the above viewpoints, and believe that the performance of carbon emissions at different stages of urbanization is different [31]. This non-linear relationship may also have regional heterogeneity, verifying the EKC hypothesis between urbanization and carbon emissions [32].

Economic growth is another major way for urbanization to increase carbon emissions, and the relationship is two-way and long-lasting. Therefore, the process of urbanization needs reasonable intervention [33]. The composition structure of carbon emissions indicates that emissions caused by energy consumption and industrial production should be given priority [12], and the potential of industrial structure adjustment to achieve carbon goals is greater than energy structure adjustment [34]. Industrial production is the main source of urban economic growth. Most of the existing studies have confirmed the positive relationship between industrialization and carbon emissions [35]. This is due to low energy efficiency, high proportion of heavy industrialization, and lack of environmental awareness in the early stages of industrialization [36]. Other studies have found an inverted U-shaped relationship between industrialization and carbon emissions [37], and it is consistent across different regions [32]. This is mainly because, with the withdrawal of heavy industry and the advancement of energy-saving technologies, the carbon emissions generated by industrialization will gradually slow down [38,39].

Technological innovation is an effective way to improve CEE [40]. Zeng et al. [41] pointed out that technological progress and foreign trade are the guarantee for improving CEE. Previous studies have shown that cleaner production technology is the key to improving environmental quality [42], but this does not mean that the relationship between technological progress and environmental quality is clear. Due to the accumulation effect and threshold effect, technological progress will not promote the improvement of production efficiency when a large amount of capital investment is required to support technological research [17,43]. In addition, after considering the bias of technological progress, Wang et al. [44] pointed out that technological progress in the industrial sector will not reduce carbon emissions, indicating that the increase of production capacity is likely to cause a rebound of carbon emissions. Similarly, there are two diametrically opposed views between the level of openness and environmental quality. "Pollution Heaven Hypothesis" believes that multinational companies will transfer severely polluting industrial chains to developing countries with relatively loose environmental regulations in order to reduce the company's environmental governance costs. The "Pollution Halo Hypothesis" believes that foreign-funded enterprises can bring advanced production technology and environmental management concepts to local enterprises, and improve local environmental and energy efficiency through knowledge spillover and technology spillover [21,45].

In summary, under the research framework of STIRPAT, a large

number of studies have conducted factor analyses of carbon emission and CEE from the three aspects of population, affluence and technology. However, many studies only use single indicators, such as the proportion of urban population to characterize the level of urbanization, and the number of scientific researchers to characterize the level of urban technology. The lack of further decomposition of key factors may be a major cause of the non-linear relationship. In addition, carbon emissions and carbon emission efficiency are not purely local environmental issues. Economic and natural factors such as foreign trade, industrial transfer, transportation, and pollution leakage may all be mechanisms for their spillover or transfer. The ignorance of spatial effects may lead to the bias of omitted variables, leading to inconsistent conclusions. In view of this, this paper will use spatial econometric models to carry out relevant empirical analysis based on the STIRPAT model.

3. Methods and data

3.1. Using nighttime light data to estimate urban carbon emissions

Nighttime light is a great proxy indicator for estimating carbon emissions [20,46]. This paper extracts the digital number values at the province and city level during 2000 to 2018 from the sensor cross-calibrated global NPP-VIIRS-like night light data [22], and then calculates the sum of digital number values (SDN). Taking into account the differences between regions, such as the difference of night lights between urban and rural areas [20], this paper uses the panel variant slope model with fixed effect to fit the relationship between carbon emissions and nighttime light [47,48]. The model is set as follows:

$$CE_{i,t} = \beta_i SDN_{i,t} + \gamma_i + \omega_t + \varepsilon_{it} \quad (1)$$

where $CE_{i,t}$ represents the provincial carbon emissions and the data source is China Carbon Emission Accounts & Datasets (CEADs) [49], $SDN_{i,t}$ represents sum of digital number, β_i is the estimated coefficient that does not change over time but changes with the individual, γ_i and ω_t represent individual fixed-effect and time fixed-effect, respectively.

3.2. Measurement of urban CEE

In general, two approaches are mainly used to evaluate the total-factor efficiency: one is parametric stochastic frontier analysis (SFA) and the other is non-parametric data envelopment analysis (DEA). The parametric method estimates the form of frontier by setting a specific production function. Limited by the form of the production function, SFA has a defect that it only applies to the case of single output, so it fails to deal with the problem of undesired output [50]. An alternative approach is to add undesired output, that is, carbon dioxide emissions, as an input to the model, which may lead to biased measurements [51]. The non-parametric DEA method avoids the problem of setting the production function, and solves the production efficiency through linear programming. Thus, the relative efficiency between decision-making units can be calculated. This paper uses the SBM-DEA model with undesired output to measure CEE [52]. The model has the characteristics of non-oriented and non-radial, that is, it is not mandatory to improve the input and output proportionally, and it can calculate the maximum possible improvement of each dimension. Therefore, the efficiency of undesired output can be measured separately [7,53]. The model can be specifically set as follows:

$$\rho^* = \min \frac{1 - \frac{1}{m} \sum_{i=1}^m \frac{s_{i0}^-}{x_{i0}}}{1 + \frac{1}{s_1 + s_2} \left(\sum_{j=1}^{s_1} \frac{s_{j0}^+}{y_{j0}} + \sum_{r=1}^{s_2} \frac{s_{r0}^b}{b_{r0}} \right)} \quad (2)$$

S.T.

$$x_0 = X\lambda + s_0^-$$

$$y_0 = Y\lambda - s_0^g$$

$$b_0 = b\lambda + s_0^b$$

$$s_0^- \geq 0, s_0^g \geq 0, s_0^b \geq 0, \lambda \geq 0$$

where x_0, y_0, b_0 represent the input, desirable output and undesirable output of the current decision-making unit, namely DMU(x_0, y_0, b_0). Each DMU has m kinds of inputs, s_1 kinds of good outputs and s_2 kinds of bad outputs. i, j, r represent their subscript indexes respectively. Vector $x \in R^m$, $y \in R^{s_1}$, $b \in R^{s_2}$. $X = [x_1, x_2, \dots, x_n] \in R^{m \times n}$ represents the input matrix. $Y = [y_1, y_2, \dots, y_n] \in R^{s_1 \times n}$ represents the desirable output matrix. $b = [b_1, b_2, \dots, b_n] \in R^{s_2 \times n}$ represents the undesirable matrix. n is the number of DMU. λ represents the weight vector of the DMU reference set. s_i^+ is the slack variable of inputs. s_j^g is the slack variable of good outputs. s_r^b is the slack variable of bad outputs.

Slack variables s_0^+, s_0^g, s_0^b represent the potential decrease of the current evaluation unit DMU input, the potential increase of the desirable output and the potential decrease of the undesirable output respectively. Therefore, according to Zhang and Choi [52], the production efficiency of undesirable output, specifically CEE in this paper, can be defined by the efficiency improvement between the actual and the frontier:

$$\begin{aligned} \text{CarbonEmissionEfficiency} &= \frac{\text{TargetEmission}}{\text{RealEmission}} \\ &= \frac{\text{RealEmission} - \text{EmissionSaving}}{\text{RealEmission}} \end{aligned} \quad (3)$$

Namely,

$$CEE = 1 - \frac{s_0^b}{b_0} \quad (4)$$

where CEE is carbon emission efficiency, s_0^b is carbon emission slack, and b_0 is real carbon emission.

The input–output variables for measuring carbon emission efficiency are as follows. Input variables include: (a) Labor force, measured by the number of employees in the city at the end of the year. (b) Capital stock, calculated by perpetual inventory method [8]. The formula is $K_{i,t} = I_{i,t} + (1 - \delta)K_{i,t-1}$, where K is the capital stock, I is the total fixed assets investment, and δ is the depreciation rate, which is set to 9.6%. The base period is set to 2003, and the annual total fixed asset investment is converted to a constant price in 2003 using the fixed asset price index of each region in each year. The base period capital stock is the total fixed asset investment in 2003. (c) Energy consumption, measured by the total annual electricity consumption. Output variables include: (d) Desirable output, which is the gross regional product (GRP) converted at constant prices in 2003. (e) Undesirable output, which is CO₂ emissions, estimated from nighttime light data.

3.3. Spatial autocorrelation test of CEE

Existing studies have shown that a competitive effect of environmental policies exists between different regions, which is manifested in that geographically adjacent regions will have a significant impact on environmental decision-making in the region. This means that environmental behavior is spatially dependent. In order to comprehensively investigate the spatial spillover effects of carbon emission efficiency, this paper conducts exploratory spatial data analysis (ESDA) of global and local spatial autocorrelation respectively.

The global spatial autocorrelation Moran's I statistic is expressed as follows:

$$\text{global Moran's I} = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\sum_{i=1}^n \sum_{j=1}^n w_{ij} \sum_{i=1}^n (y_i - \bar{y})^2} \quad (5)$$

where n is the number of space units, w_{ij} is the element in the spatial

weight matrix, y is the CEE columns and $\bar{y}$ is the arithmetic mean respectively. Moran's I is restricted to the range of [-1, 1]. When Moran's I > 0, it represents the positive spatial autocorrelation, that is, spatial clustering. Moran's I < 0 indicates negative spatial autocorrelation, that is, spatial dispersion. Moran's I = 0 means CEE is spatially distributed at random.

Global Moran's I reflects the overall spatial correlation of CEE, but some local features may be ignored. For example, the positive correlation feature of some areas will cancel out the negative correlation of neighboring areas [18]. Therefore, Anselin [54] proposed the local indicator of spatial association (LISA) to decompose the global Moran's I. Specifically,

$$I_i = \frac{(y_i - \bar{y})}{S_i^2} \sum_{j \neq i}^n w_{ij} (y_j - \bar{y}) \quad (6)$$

where,

$$S_i^2 = \frac{1}{n-1} \sum_{j \neq i}^n w_{ij} y_j^2 - \bar{y}^2 \quad (7)$$

When $I_i > 0$, it means that high-CEE areas are surrounded by high-CEE areas (H-H type) or low-CEE areas are surrounded by low-CEE areas (L-L type). When $I_i < 0$, it means that high-CEE areas are surrounded by low-CEE areas (H-L type) or low-CEE areas are surrounded by high-CEE areas (L-H type).

In order to enhance the robustness of the results, this paper constructs four spatial weight matrices. The first is the contiguity matrix (Rook) based on geographic proximity, that is, if two cities are adjacent, w_{ij} is set to 1, otherwise it is 0. The second is the inverse distance weight matrix (Dist), w_{ij} represents the inverse of the Euclidean distance between city i and city j . The third and fourth are K-nearest neighbors spatial weight matrices. This paper selects K4 and K5 respectively. It represents that each city has 4 or 5 nearest neighboring cities, which will have spatial spillover effects on it. In addition, in order to make the sum of the influences of all other spatial units to be the same for the one unit, this paper performs row normalization on the spatial weight matrix.

3.4. Spatial econometric models

Based on the STIRPAT model and the spatial econometric model, this paper constructs a theoretical framework for the analysis of factors affecting CEE. The STIRPAT model is the stochastic form of IPAT model [55]:

$$I = aP^b A^c T^d \varepsilon \quad (8)$$

Where I, P, A , and T are the indicator variables of environmental quality, population, affluence, and technology. ε is the random error term. b, c , and d are the elastic coefficients between P, A , and T respectively, which makes the model can be used to analyze the non-proportional effects of different factors on the environment. After taking the natural logarithm on both sides of the equation, the elastic coefficients can be used as parameters for estimation.

This paper adopts spatial Durbin error model (SDEM) to reflect the spatial dependence between cities. The model takes into account the spatial dependence of the disturbance term and the local spatial spillover effect. Specifically, the benchmark SDEM is set as follows:

$$\begin{aligned} CEE_{i,t} &= \alpha_0 + \alpha X_{i,t} + \theta \sum_j w_{ij} X_{j,t} + \mu_i + \omega_t + u_{i,t} \\ u_{i,t} &= \lambda \sum_j w_{ij} u_{j,t} + \varepsilon_{i,t} \end{aligned} \quad (9)$$

where α_0 is a constant term. X is a set of explanatory variables of influencing factors and α is the estimated parameter vector. $\sum_j w_{ij} X_{j,t}$ denotes the lagged dependent variable. θ reflects the spatial impacts

from explanatory variables in neighboring cities. $\sum_j w_{ij} u_{i,t}$ represents the spatially autocorrelated error term. λ is the spatial autocorrelation coefficient. μ_i and ω_t are added to control the spatial fixed-effect and time fixed-effect.

The explanatory variable X includes: (a) Urban population characteristics: Population size (*lnpop*); Population density (*lnpopd*); Human capital level (*lnedu*), measured by the number of full-time college teachers per 10,000 people. (b) Urban economic development: Per Capita Gross Regional Product (*lngrpc*); Industrial structure (*sec*), measured by the proportion of the added value of the secondary industry in GRP; Degree of manufacturing industry agglomeration (*magg*), measured by the method of location entropy. (c) Technical development level: Clean production level (*lnele*), measured by industrial power consumption level in unit industrial output; R&D investment (*Tech*), which is the proportion of government expenditure on supporting science and technology undertakings in the fiscal revenue; Foreign direct investment (*lnfdi*), refers to the proportion of actually utilized FDI in GRP.

According to the theory of economies of scale, location theory, and new economic geography, the spatial aggregation of economic activities can improve production efficiency through cost savings in production and logistics, infrastructure sharing, technology and knowledge spillovers, etc. To this end, this paper considers the interactive effects of regional openness and industrial collaboration. Apart from manufacturing industry agglomeration (*magg*), producer service industry agglomeration indicator (*sagg*) is further added, and their coordination agglomeration indicator (*coagg*) is examined. *magg* and *sagg* are measured by the method of location entropy. This method eliminates the difference in the scale of factors between regions and can reflect the geographical distribution of factors. It is expressed as follows:

$$agg_i = \left(\frac{q_i}{\sum_i q_i} \right) / \left(\frac{Q_i}{\sum_i Q_i} \right) \quad (10)$$

where *agg* represents the location entropy of the industry. The larger the value, the higher the industrial agglomeration. q_i represents the number of employees in the manufacturing industry (or producer service industry) in city i , and Q_i represents the total number of employees. Further, the co-aggregation degree of *magg* and *sagg* is calculated by:

$$coagg = 1 - \frac{|magg - sagg|}{magg + sagg} + magg + sagg \quad (11)$$

Considering the availability of data, this paper finally selects 282 cities in China from 2004 to 2018 as the research sample, and the missing data in a few years are supplemented by linear interpolation. All monetary variables are converted to constant prices in 2003, and non-percentage variables are taken into the econometric model after taking the natural logarithm. Unless otherwise specified, data in this paper are obtained from China City Statistical Yearbook and China Statistical Yearbook. The descriptive statistics of the main variables are shown in Table 1.

Table 1
Descriptive statistics for variables.

Variable	Obs	Mean	SD	Min	Median	Max
CEE	4230	0.541	0.183	0.060	0.509	1.000
lnpop	4230	5.862	0.694	2.819	5.911	8.133
lnpopd	4230	5.725	0.918	1.547	5.872	7.887
lnedu	4230	1.850	0.918	0.000	1.696	4.436
lngrpc	4230	9.999	0.809	7.631	9.949	12.817
sec	4230	0.482	0.110	0.090	0.484	0.910
magg	4230	0.902	0.577	0.023	0.787	4.933
tech	4230	0.058	0.142	0.000	0.025	2.800
lnele	4230	6.610	0.980	2.112	6.643	10.841
lnfdi	4230	9.219	2.290	0.000	9.433	14.609

4. Results and discussion

4.1. Results of carbon emissions and CEE

When forecasting carbon emissions for a long time series, it is necessary to check the stationarity of variables. This paper adopts four types of panel unit root tests including LLC test, IPS test, Fisher-ADF test, Fisher-PP test. The results are shown in Table 2. The first-difference series of these variables are stationary, and at the 1% significance level, the null hypothesis of the existence of a unit root is rejected. Since these variables are integrated process of order one, the panel cointegration tests based on the Pedroni residual method are used. The Pedroni test allows different panel units to have their own cointegration vectors and residual autoregressive coefficients, and adds a time trend term. The results are shown in Table 3. All variables are cointegrated and the error term follows a stationary process. Therefore, the regression analysis is not spurious, and there is a long-term relationship between carbon emissions and SDN.

As is shown in Table 4, this paper attempts four models to fit the relationship between SDN and carbon emission. The four models are variant slope model, variant slope model with time fixed-effect, variant slope model with individual fixed-effect and variant slope model with two-way fixed effect. The two-way fixed effects model was the best fit, with adjusted R^2 showing that SDN explained 96.43% of the variance in CO_2 . Therefore, the model specified in Eq. (1) is used to predict carbon emissions at the city level. The prediction results from 2004 to 2018 are shown in Fig. 1. The longitude of 282 cities is arranged from small to large on the horizontal axis, so each band reflects the changing trend of carbon emissions from western to eastern regions. The western region has the least carbon emissions, followed by the northeastern region, and the central and eastern cities have higher carbon emissions. This distribution is in line with the characteristics of China's regional economic development and industrial activities. In terms of time dimension, there was a significant leap forward growth in urban carbon emissions in 2005 and 2010, which coincided with the junction of the two "Five-Year Plans" (FYP). Under the pressure of assessment or the introduction of new preferential policies, economic activities will increase significantly, resulting in a large increase in carbon emissions.

Fig. 2 shows the regional and national averages of CEE measurement results. Generally, the carbon emission efficiency of various regions showed a positive U-shape change trend, reaching the lowest point in 2012, and then gradually rising. It is consistent with the results calculated by Yu and Zhang using SBM and DDF methods [7], which verifies that the method of using nighttime light data to estimate carbon emissions is feasible. According to Table 1, the average CEE of the whole sample is 0.541, which means there is still 45.9% room for improvement. This U-shaped change trend supports the conclusion of EKC curve from the perspective of production efficiency, that is, in the early stage the significant increase of economic activities will lead to the significant deterioration of environmental quality, but eventually lead to the improvement with the further development of economy. However, CEE has declined to a certain extent in 2018, which is related to the overall rebound in carbon emissions. Galik [56] points out that carbon emissions in China rebounded from 2017 to 2019, ending the steady decline in 2013–2016.

After crossing the lowest point, CEE has been significantly improved from 2012 to 2013, which is mainly due to the great changes in the environmental policy at that time, especially the policy on carbon emissions. In November 2012, the 18th National Congress of the Communist Party of China made a decision of vigorously promoting the construction of ecological civilization, established the strategic position of the Scientific Outlook on Development. The construction of China's ecological civilization entered a new historical period. Subsequently, in view of the fact that the first batch of low-carbon city pilot areas in 2010 were provincial-level or larger central cities, the policy implementation effect was not good. In order to promote the sinking of pilot policies, the

Table 2

Panel unit root test results.

	Variable	LLC	IPS	Fisher-ADF	Fisher-PP
Level	CO ₂	-2.7979***	-1.5156*	150.4876***	53.8622
	SDN	-2.5500***	-6.4378***	75.4726*	137.3117***
First difference	CO ₂	-4.3010***	-9.7107***	223.4299***	341.4660***
	SDN	-12.1024***	-13.8924***	250.5605***	965.2306***

Note: *** and * denote statistical significance at the 1% and 10% levels.

Table 3

Panel cointegration test results.

	Panel PP test	Panel ADF test	Group PP test	Group ADF test
Statistic	-3.3254***	-3.3665***	-2.5980***	-2.5955***

Note: *** denotes statistical significance at the 1% level.

Table 4

Variant slope model results.

	(1)	(2)	(3)	(4)
	CO ₂	CO ₂	CO ₂	CO ₂
SDN	0.000585*** (0.000040)	0.000472*** (0.000000)	0.000330*** (0.000079)	0.000210*** (0.000070)
Constant	106.4642*** (14.588572)	147.8463*** (0.000000)	78.9735*** (9.501404)	96.5954*** (7.944178)
Area Fixed-effect	No	Yes	No	Yes
Time Fixed-effect	No	No	Yes	Yes
Observations	570	570	570	570
Adjusted R ²	0.8741	0.9116	0.9289	0.9643

Note: Standard errors in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

National Development and Reform Commission (NDRC) organized the second batch of national low-carbon provinces and cities at the end of 2012, mainly including Beijing, Shanghai, Hainan Province and 26 prefecture-level cities¹. Finally, in 2011, the NDRC issued the *Notice on Carrying out the Pilot Work of Carbon Emission Trading*² to promote the use of market mechanisms in seven provinces and cities including Beijing, Tianjin, etc. However, the overall pilot work was not fully implemented until 2013. It can be seen that the implementation of these policies has played a pivotal role in the improvement of CEE.

4.2. Spatial agglomeration of CEE

Taking rook matrix as an example, Fig. 3 shows the Moran scatter plot of urban CEE in 2006, 2011 and 2016. The horizontal axis represents decentralized CEE, and the vertical axis represents the spatial lag value of CEE. It can be seen that most cities are located in the first and third quadrants, indicating that there is a positive spatial autocorrelation characteristic of H-H type or L-L type in CEE.

Fig. 4 shows the dynamic evolution characteristics of local cluster of CEE in a certain years. The red areas represent the H-H type agglomeration areas at the 5% significance level, and the blue represents the L-L type. It can be seen that the H-H type cities in 2004 were mainly distributed in the Northeast, the central and western regions of Inner Mongolia, Guangxi, and the junction of Fujian and Guangdong. The northwestern region has always maintained a high degree of clustering, but the number of clustered cities has gradually decreased. The high concentration of the Northeast region and the junction border of Fujian

and Guangdong gradually dispersed until 2017 began to withdraw from the H-H type areas. Another high concentration areas in Guangxi have gradually shifted to central region such as Hunan Province. Unlike the discrete distribution of high-high accumulation areas, the L-L type cities are mainly concentrated in patches. The central region includes south-western Shanxi, southern Hebei, most of Henan, northern Anhui, and northern Hubei, which are always in low CEE cluster areas. Northern Jiangsu and Southern Shandong have largely withdrawn from low-efficiency clusters after 2008. Sichuan-Chongqing region began to appear low-aggregation areas since 2013 and expanded year by year. On the whole, the L-L type agglomeration areas show a trend of migration from the North China Plain to the Southwest.

4.3. Empirical results of spatial econometric models

Both global and local autocorrelation analysis confirmed the spatial clustering characteristics of CEE. This proves that the model setting should consider the spatial correlation of variables, so the spatial econometric model is used in this paper. Before parameter estimation, LM test is used to select empirical model between SLM and SEM. The results are shown in Table 5. The LM statistics of the two show the same significance. Therefore, robust LM test is further used to determine the form of model. The results show that under the four spatial weight matrices, the robust LM statistic for the spatial error model is significant at the 1% level, while the robust LM statistic for the spatial lag model cannot reject the null hypothesis that there is no lag term except for K5 weight matrix. Thus basically it shows the SEM model is better than the SLM model. According to Elhorst [57], although the LM test rejects the non-spatial model in favor of the spatial lag model or the spatial error model, the choice of either model should still be done with care. Therefore, based on the consideration of the robustness of the results, the spillover effect of explanatory variables is further considered on the basis of the SEM model, that is, the SDEM model is adopted.

Table 6 shows the benchmark regression results of SDEM based on Eq. (9). Under the four weight matrices, Moran's I tests for regression error are all not significant, indicating that the spatial dependence in the error term is well controlled. It can be seen that the coefficients and directions of the variables under different spatial weight matrices are basically the same, and only a few variables have changed the degree of significance. Note that λ under the Dist matrix is significantly larger than the other models. This is mainly due to the fact that the inverse distance weight matrix incorporates the spatial dependencies between any two regions, while the other three weight matrices only consider the local spatial dependencies. According to Elhorst [57], this paper also reports two goodness-of-fit measures (R^2 and corr^2) for each model. Table 7 shows the robustness test results of SDEM. Considering the sensitivity of the spatial econometric results to different spatial weight matrices, this paper further adopts the economic distance matrix and the highway driving distance matrix, as shown in the Columns (1)-(2) respectively. The economic distance matrix is constructed from the GRP of each city and the highway distance between the two cities is optimally planned by Gaode map. The industrial structure is still the main factor for the reduction of carbon emission efficiency, but under the economic distance matrix, the coefficient of *magg* becomes insignificant. It may be that the economic distance matrix and the highway driving distance matrix are not purely geographical distances but belong to the socio-

¹ National Development and Reform Commission (NDRC), China, http://www.ncsc.org.cn/SY/dtsdysf/202003/t20200319_769716.shtml

² National Development and Reform Commission (NDRC), China, <https://zfxgk.ndrc.gov.cn/web/iteminfo.jsp?id=1349>

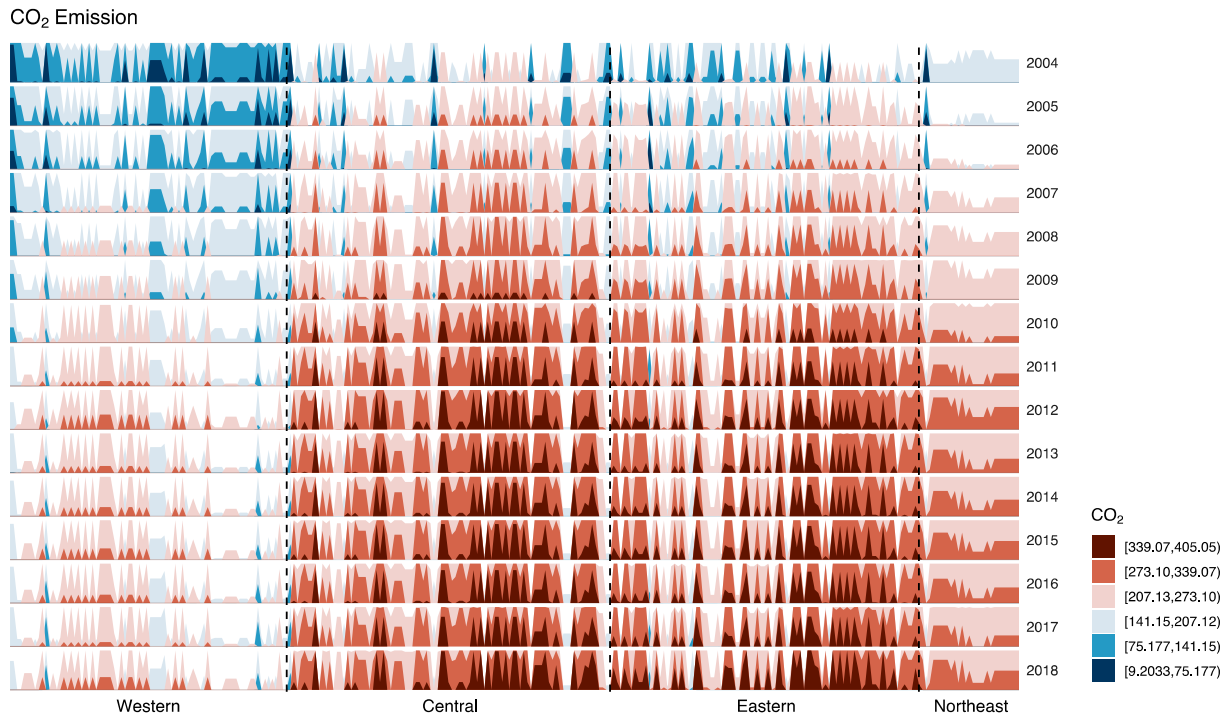


Fig. 1. The temporal and spatial distribution of carbon emissions.

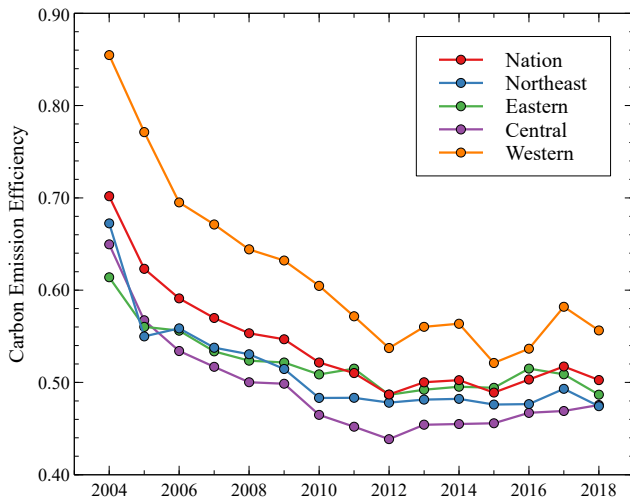


Fig. 2. The trend of CEE in different regions: 2004–2018.

economic weight matrix which will have a certain endogenous impact on the model estimation. The log likelihood values under both weight matrices are greater than the values under the geographic weight matrices, indicating that the models in Table 6 have a higher goodness of fit. In addition, considering that the raw nighttime time data may have the characteristics of temporal discontinuities and outliers in high latitudes, Column (3) removes the samples after 2013 and Column (4) excludes samples of cities with latitudes higher than 45 degrees. There is no significant difference in the conclusions, and *sec* and *magg* are still the main factors for the reduction of carbon emission efficiency. Therefore, the following analysis in this paper is still based on the four spatial weight matrices shown in Table 6.

First of all, the three variables that characterize the urban population act in different directions. *lnpop* had a significant positive effect on CEE, while *lnpopd* and *lnedu* were negative but not significant. The expansion

of the urban population has significantly improved the efficiency of carbon emissions. It reflects that during the large migration of rural to urban, the intensive effect of infrastructure sharing has played a positive role. For example, the utilization rate of public transportation has been increased, and the cost of pollution control and emission reduction can be shared together. However, with the further gathering of the population, some urban diseases have also begun to appear. In the later stages of urbanization, the improvement of people's living standards is no longer satisfied with compact lifestyles, and higher requirements are put forward for personal living space, housing heating and cooling, and private car travelling. As a result, the increase in personal energy consumption has led to an increase in carbon dioxide emissions. Blind expansion of cities lower the land use efficiency, road congestion leads to insufficient fuel combustion of vehicles, and saturation of pollution control facilities cannot afford environmental pollution. These are ultimately reflected in the negative impact of population density on CEE, which is worth noting although it is not significant at this stage.

Secondly, the three decomposition variables of economic development level are all significant at the 1% level. The coefficient of *lngrpc* is significantly positive, indicating that most cities attach importance to the construction of ecological civilization and green modernization to achieve urban environmental transformation. After adding the quadratic term of *lngrpc*, it is found that the coefficient of *lngrpc*² is significant and positive, and the coefficient of *lngrpc* is negative, indicating that there is a U-shaped relationship between CEE and *lngrpc*. It provides new evidence for the EKC curve of carbon emissions from an efficiency perspective. The coefficient of *sec* is significantly negative. This is not difficult to understand, because the secondary industry is the main source of carbon emissions. In view of the pivotal position of manufacturing in the secondary industry, this paper further considers the impact of manufacturing industry agglomeration. Unsurprisingly, *magg* also led to a decrease in CEE. But the spatial spillover effects of *sec* and *magg* show diametrically opposite situations. The spatial coefficient of *W-sec* is significantly negative, while *W-magg* is significantly positive, indicating that the industrial structure has a further negative impact on the CEE of neighboring regions, while the impact of manufacturing agglomeration on the CEE of neighboring regions is positive. This

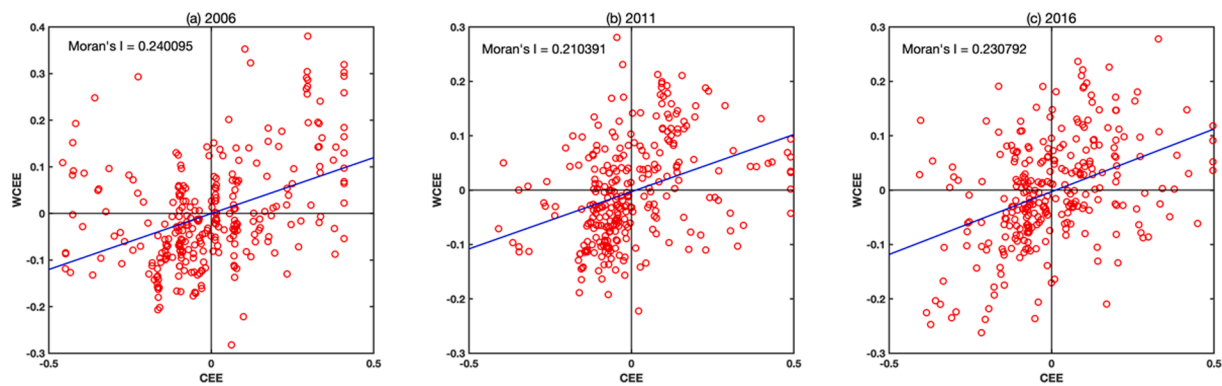


Fig. 3. Moran scatter plot under Rook weight matrix: (a) 2006 (b) 2011 (c) 2016.

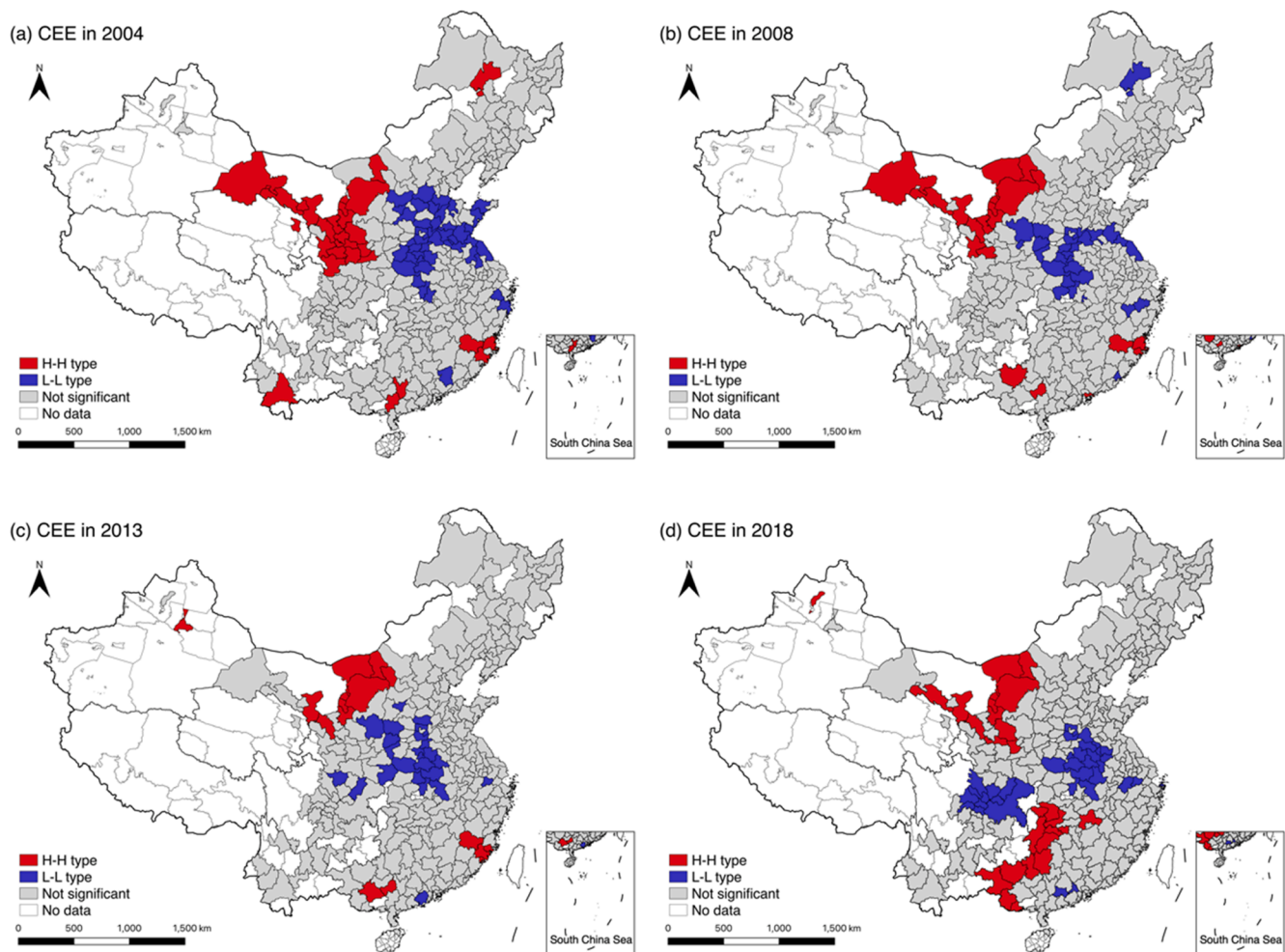


Fig. 4. LISA agglomeration maps of CEE: (a) 2004, (b) 2008, (c) 2013, (d) 2018.

reflects the similarity between regions in the environmental impact of industrial structures. The increase in the proportion of the secondary industry may be realized by the transfer of high-energy-consuming industries in the surrounding areas or the unfavorable trade division of labor. In the process, the existence of carbon leakage has reduced the carbon emission efficiency of both the local and neighboring regions. However, the environmental cost of manufacturing agglomeration has a warning effect, and the extensive development method may not be imitated by neighboring regions.

Finally, technological progress will affect CEE in two ways. When

R&D expenditure is widely invested in low-carbon production technologies, it can bring green growth effects to environmental quality. However, technological progress may also bring about an energy rebound effect, that is, R&D investment simply improves production efficiency, and the reduction of production costs enables enterprises to increase larger production capacity, which leads to higher carbon emissions. The coefficient of *tech* is positive, indicating that the green growth effect brought by technology investment is greater than its energy rebound effect. The coefficient of *lnle* is significantly positive, that is, the higher the electricity power consumption level per industrial output,

Table 5
LM test of spatial panel model.

LM test	Rook	Dist	K4	K5
LM: no lag	70.5051*** (0.000)	98.8233*** (0.000)	104.2628*** (0.000)	108.4915*** (0.000)
Robust LM: no lag	0.7866 (0.375)	0.6581 (0.417)	2.6413 (0.104)	4.4333** (0.035)
LM: no error	82.1463*** (0.000)	132.5663*** (0.000)	111.5472*** (0.000)	113.0210*** (0.000)
Robust LM: no error	12.4279*** (0.000)	34.4011*** (0.000)	9.9257*** (0.002)	8.9627*** (0.003)

Note: p value in parenthesis. *** and ** denote statistical significance at the 1% and 5% levels.

the higher the regional CEE, indicating the importance of cleaner production. The new energy industry has been deployed on a large scale in China since 2006. At present, China has developed the world's largest hydropower, solar photovoltaic and wind energy. With the implementation of China's west-to-east power transmission project and the establishment of UHV transmission lines, more and more of China's power composition comes from clean energy, which is an important and

positive factor on CEE.

In order to promote the transformation and upgrading of urban industrial structure, this paper further studies the mechanism of mitigating the reduction of carbon emission efficiency from the perspective of industrial agglomeration. Table 8 shows the effects of industrial collaboration. The productive service industry provides necessary capital turnover and supporting services for the purchasing, production and sales of manufacturing enterprises, which can reduce the costs of manufacturing enterprises and improve operation efficiency. Therefore, the collaborative aggregation of the producer service industry and the manufacturing industry has a significant value-added effect. The results of Table 8 also confirm this point. The agglomeration indicator *sagg* of the producer service industry was added. The results show that adding 1 unit of *sagg* will significantly increase CEE by 3%. Columns (5)–(8) in Table 8 shows the synergistic aggregation of the producer service industry and the manufacturing industry under different spatial weight matrices. The results show that the coefficients of *coagg* are all significantly positive, indicating that the development of supporting industries deeply integrated with the manufacturing industry can produce a benign synergistic effect on CEE. The establishment of a complete industrial ecosystem in the city can drive the high-quality development of the

Table 6
Regression results of SDEM.

Variable	(1) Rook	(2) Dist	(3) K4	(4) K5	(5) Rook
lnpop	0.1567*** (4.351)	0.1506*** (4.283)	0.1201*** (3.460)	0.1380*** (3.970)	0.1303*** (3.531)
lnpopd	−0.0330 (−1.074)	−0.0111 (−0.364)	−0.0165 (−0.540)	−0.0172 (−0.563)	−0.0259 (−0.826)
lnedu	−0.0026 (−0.284)	0.0012 (0.136)	−0.0019 (−0.204)	−0.0020 (−0.217)	−0.0083 (−0.888)
lngrpc	0.0983*** (5.768)	0.0830*** (4.922)	0.0748*** (4.364)	0.0818*** (4.795)	−0.7945*** (−10.075)
lngrpc ²					0.0444*** (11.612)
sec	−0.1209*** (−3.372)	−0.1215*** (−3.464)	−0.1345*** (−3.786)	−0.1225*** (−3.455)	−0.0090 (−0.237)
magg	−0.0185*** (−3.095)	−0.0193*** (−3.307)	−0.0204*** (−3.355)	−0.0205*** (−3.347)	−0.0202*** (−3.300)
lnele	0.0195*** (5.465)	0.0204*** (5.731)	0.0186*** (5.220)	0.0182*** (5.131)	0.0251*** (6.853)
tech	0.1860*** (7.949)	0.1835*** (8.066)	0.1686*** (7.142)	0.1655*** (7.003)	0.0902*** (3.566)
lnfdi	−0.0001 (−0.052)	−0.0016 (−0.821)	0.0002 (0.115)	0.0000 (0.011)	0.0013 (0.642)
W-lnpop	−0.2355*** (−3.194)	−0.3105*** (−2.857)	−0.2026*** (−3.249)	−0.2307*** (−3.405)	−0.2794*** (−3.748)
W-lnpopd	0.0440 (0.647)	−0.0430 (−0.429)	0.0626 (1.109)	0.0880 (1.404)	0.0907 (1.319)
W-lnele	−0.0506** (−2.209)	−0.0349 (−1.175)	−0.0856*** (−4.345)	−0.0980*** (−4.367)	−0.0460** (−1.987)
W-lngrpc	−0.0610** (−1.960)	0.0251 (0.588)	0.0171 (0.621)	0.0229 (0.804)	−0.0763** (−2.426)
W-sec	−0.2608*** (−3.512)	−0.5677*** (−5.422)	−0.2071*** (−3.131)	−0.3201*** (−4.486)	−0.1425* (−1.885)
W-magg	0.0747*** (6.753)	0.1060*** (7.749)	0.0659*** (7.042)	0.0677*** (6.702)	0.0739*** (6.616)
W-lnele	0.0038 (0.467)	−0.0107 (−0.881)	0.0143* (1.909)	0.0210** (2.574)	0.0026 (0.326)
W-tech	0.0259 (0.597)	0.0172 (0.279)	0.0774* (1.939)	0.0941** (2.287)	0.0050 (0.114)
W-lnfdi	0.0068* (1.684)	0.0133** (2.310)	0.0005 (0.142)	0.0029 (0.735)	0.0113*** (2.762)
lambda	0.1730*** (7.127)	0.3110*** (9.904)	0.1860*** (8.824)	0.1870*** (8.031)	0.1554*** (6.347)
error moran's I	−0.0164 (−0.479)	−0.0068 (−0.137)	−0.0154 (−0.515)	−0.0162 (−0.489)	−0.0176 (−0.512)
log likelihood	3413.01	3443.70	3429.97	3437.15	3483.46
R ²	0.6481	0.6500	0.6490	0.6513	0.6611
corr ²	0.0637	0.0685	0.0661	0.0722	0.0982
N	4230	4230	4230	4230	4230

Note: t-statistics in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7
Robustness test results of SDEM.

Variable	(1) W_econ	(2) W_road	(3) Rook	(4) Rook
lnpop	0.0611* (1.847)	0.1498*** (4.300)	0.1136** (2.425)	0.1858*** (5.007)
lnpopd	−0.0394 (−1.284)	0.0157 (0.512)	−0.0108 (−0.322)	−0.0455 (−1.454)
lnedu	−0.0021 (−0.234)	−0.0001 (−0.013)	0.0174 (1.482)	−0.0009 (−0.095)
lngrpc	0.0554*** (3.703)	0.1032*** (6.115)	0.0344 (1.598)	0.1259*** (6.812)
sec	−0.1737*** (−5.100)	−0.0989*** (−2.809)	−0.1015** (−2.474)	−0.1542*** (−4.084)
magg	0.0030 (0.550)	−0.0181*** (−3.037)	−0.0486*** (−4.584)	−0.0177*** (−2.928)
lnele	0.0180*** (5.015)	0.0184*** (5.215)	0.1651*** (7.304)	0.1729*** (7.144)
tech	0.1400*** (6.485)	0.1603*** (6.968)	0.0001 (0.023)	0.0215*** (5.839)
lnfdi	0.0001 (0.027)	−0.0010 (−0.497)	−0.0063** (−2.214)	−0.0013 (−0.642)
W-lnpop	0.1942** (2.025)	−1.3879*** (−2.922)	−0.1737 (−1.616)	−0.2164*** (−2.814)
W-lnpopd	0.0149 (0.195)	0.6597 (1.424)	0.0410 (0.529)	0.0853 (1.231)
W-lnedu	0.0216 (0.826)	−0.7768*** (−4.476)	−0.0291 (−1.020)	−0.0584** (−2.420)
W-lngrpc	−0.0311 (−0.718)	0.0780 (0.496)	0.0094 (0.216)	−0.0849** (−2.540)
W-sec	−0.5564*** (−5.436)	−2.3347*** (−4.722)	−0.2280*** (−2.634)	−0.2318*** (−2.992)
W-magg	0.0180 (1.069)	0.4063*** (6.972)	0.0320 (1.396)	0.0739*** (6.671)
W-lnele	−0.0410*** (−4.155)	0.0478 (0.746)	0.0084 (0.203)	0.0291 (0.652)
W-tech	0.2344*** (4.473)	0.6437*** (2.826)	0.0107 (0.893)	0.0085 (1.025)
W-lnfdi	−0.0068 (−1.386)	0.0667*** (2.761)	0.0051 (0.877)	0.0068 (1.636)
lambda	0.2820*** (10.097)	0.4910*** (6.028)	0.1630*** (5.456)	0.1610*** (6.423)
log likelihood	3449.40	3444.65	2528.77	3247.75
R ²	0.6500	0.6563	0.7195	0.6532
corr ²	0.0688	0.0854	0.0499	0.0684
N	4230	4230	2820	4020

Note: t-statistics in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

regional economy. In addition, the spatial lag terms of *sagg* and *coagg* are both positive, reflecting the demonstration effect of producer service agglomeration and synergistic agglomeration. Therefore, the agglomeration of producer services should be cultivated and strengthened, the coupling and coordination between producer services and manufacturing should be optimized, and the synergistic effect of industrial agglomeration on the growth of urban carbon emission efficiency should be better played.

5. Conclusions

This paper uses panel data of 282 cities in China from 2004 to 2018 to study the influencing factors of CEE. The variant slope model is used to estimate the city's carbon emissions based on night light data, and the urban CEE is measured. The results show that carbon emission sources are mainly distributed in central and eastern cities, and there is a significant increase after 2010. In the time dimension, the carbon emission efficiency shows a U-shaped change trend, and this trend is confirmed in the EKC test related to per capita GRP.

In the spatial dimension, the results of the global spatial autocorrelation test show that urban CEE has significant and positive spatial spillover effects. Local spatial association test indicates that the H-H type areas are scattered in the northwest and southwest of China, and the aggregation areas in the northwest are gradually fragmented. The L-L type areas migrated from the eastern coastal area to the southwest area, and eventually formed two low-efficiency centers in the Central Plains urban agglomeration and the Sichuan-Chongqing urban agglomeration. It shows that there likely exists a phenomenon of “race to the bottom” in CEE between regions, that is, in order to improve urban competitiveness and obtain more production capital, cities may lower the environmental regulatory standards. Since this kind of environmental decision-making is likely to have a demonstration effect, it leads to agglomeration of low-efficiency areas.

The spatial Durbin error model is further used to analyze the influencing factors of urban carbon emissions efficiency in this paper. The results of the spatial econometric model show that the expansion of urban population has a positive impact on CEE, while population density and human capital levels have a negative but insignificant impact, indicating that the intensive effect of population concentration on CEE is greater than the crowding effect at this stage. With the deepening of the urbanization process, densely populated areas will generate a large amount of demand for housing, transportation and living energy. Therefore, on the one hand, it is necessary to reasonably guide the

Table 8
The effect of industrial collaboration.

Variable	(1) Rook	(2) Dist	(3) K4	(4) K5	(5) Rook	(6) Dist	(7) K4	(8) K5
sagg	0.0309*** (5.688)	0.0286*** (5.310)	0.0283*** (5.219)	0.0284*** (5.214)				
W-sagg	0.0156 (1.210)	0.0414*** (3.082)	0.0378*** (3.335)	0.0413*** (3.213)				
coagg					0.0323*** (5.995)	0.0294*** (5.512)	0.0281*** (5.210)	0.0289*** (5.327)
W-coagg					0.0204* (1.667)	0.0454*** (3.438)	0.0420*** (3.995)	0.0429*** (3.600)
magg	−0.0315*** (−4.952)	−0.0335*** (−5.331)	−0.0341*** (−5.300)	−0.0335*** (−5.216)	−0.0639*** (−6.808)	−0.0629*** (−6.747)	−0.0619*** (−6.579)	−0.0620*** (−6.604)
W-magg	0.0573*** (4.441)	0.0774*** (5.267)	0.0408*** (3.794)	0.0376*** (3.108)	0.0346 (1.638)	0.0318 (1.428)	−0.0013 (−0.075)	−0.0047 (−0.233)
lambda	0.1690*** (6.949)	0.3010*** (9.528)	0.1730*** (8.143)	0.1730*** (7.362)	0.1660*** (6.816)	0.3000*** (9.492)	0.1660*** (7.781)	0.1670*** (7.082)
controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
log likelihood	3430.60	3461.56	3451.00	3458.38	3433.76	3464.41	3454.78	3461.43
R ²	0.6513	0.6532	0.6535	0.6555	0.6520	0.6538	0.6545	0.6562
corr ²	0.0722	0.0772	0.0780	0.0832	0.0739	0.0786	0.0806	0.0850
N	4230	4230	4230	4230	4230	4230	4230	4230

Note: t-statistics in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

process of population urbanization, strengthen regional opening, and give full play to the intensive effect of population size. On the other hand, expansion of the urban infrastructure construction should be in an orderly manner, and be alert to the crowding effect caused by excessive population growth.

Green economic development is the goal of many cities. This paper verifies the green growth effect of urban per capita GRP, but industrial scale and agglomeration adversely affect CEE. Industrial scale had the same adverse impact on local and neighboring CEE, while manufacturing agglomeration mainly had a negative impact on local CEE, but had a warning effect on neighboring regions. Fortunately, this paper further finds that the synergistic agglomeration of manufacturing and producer services industries can effectively improve CEE. It shows that strengthening the regional and industrial cooperation of production activities is an effective way for cities to improve the efficiency of green production, which provides a new decision basis for the city's industrial layout and structural adjustment.

Technological progress is another important way to improve environmental quality. Technical variables are decomposed into input variables (R&D investment and foreign investment) and performance variables (electric energy consumption). The results show that R&D investment and electricity consumption positively affect carbon emission efficiency, but the effect of foreign investment is not significant. It verifies that R&D investment and cleaner production technology have played a green growth effect, and shows the importance of the guidance of green technology progress in the innovation system. In addition, prefecture-level cities also need to improve the environmental access conditions for FDI projects to prevent disorderly competition and energy rebound effects. Ensure that the technological advantages and spillover effects of foreign capital in environmental governance can be actively brought into play.

The achievement of carbon peak and carbon neutrality goals is a systematic project that requires active cooperation from various aspects of the economy and society. The regional agglomeration of CEE means that the emission reduction and efficiency improvement efforts of a single city may be futile due to the mutual influence between regions. The importance of regional cooperation is highlighted here. Therefore, each individual city must reach a consensus on the environmental benefits as well as responsibility boundaries of an overall subject. Form environmental regulations, ecological standards and compensation mechanisms for regional cooperation. Under the overall environmental constraints, jointly plan and implement energy-saving and emission-reduction programs, give full play to the positive role of industrial and regional collaboration, force urban green transformation, economic structure optimization and technological innovation breakthroughs, so as to minimize the cost of individual low-carbon development in the region.

CRedit authorship contribution statement

Guochang Fang: Conceptualization, Methodology, Writing – original draft, Visualization, Funding acquisition. **Zhengye Gao:** Data curation, Investigation, Writing – original draft, Software, Visualization. **Lixin Tian:** Resources, Validation, Funding acquisition. **Min Fu:** Data curation, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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